

A **rational function** is a quotient of polynomials:

$$f(x) = \frac{P(x)}{Q(x)}, \quad Q(x) \neq 0.$$

Its domain includes every real number except the real zeros of the *original* denominator.

1. Factor the denominator and find its real zeros.
2. Record the excluded values before canceling common factors.
3. Simplify when possible, while keeping the original exclusions.

If a common factor cancels and the reduced denominator is nonzero there, the graph has a **hole**. Use the reduced formula to find its height.

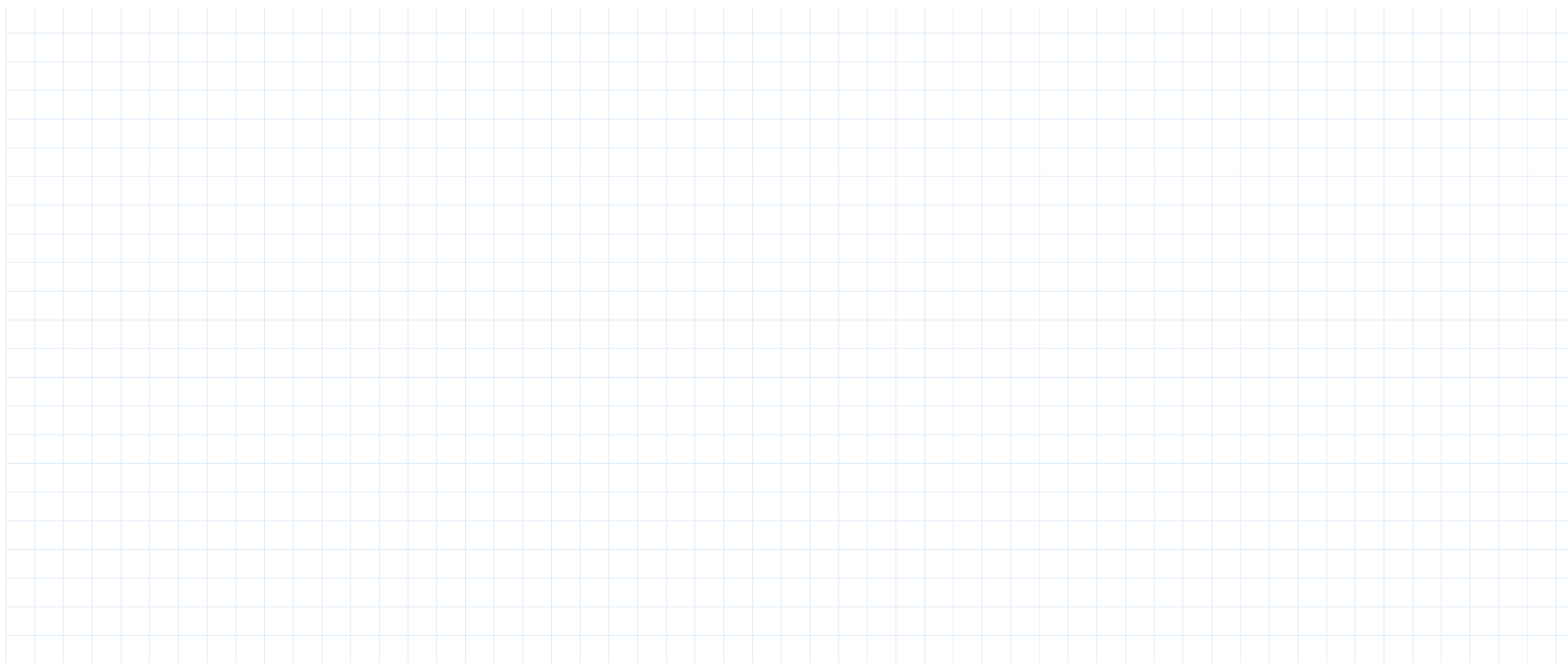
Find the Domain

Section 3.4A

Find the domain of

$$f(x) = \frac{x + 4}{x^2 + x - 6}.$$

Show the factorization and write the domain in interval notation.

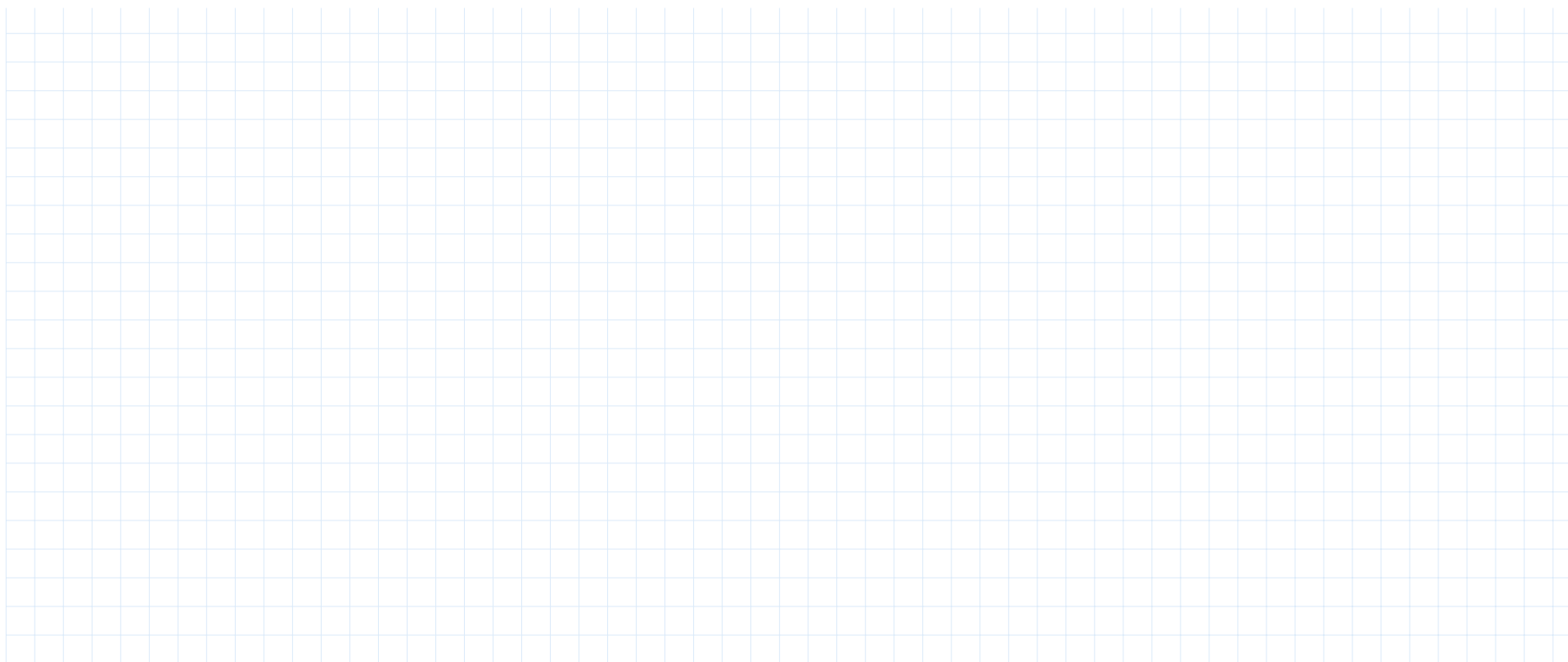


Special Domain Cases

Find the domain of each function. Explain any excluded values.

(a) $g(x) = \frac{2x - 5}{4}$

(b) $h(x) = \frac{3}{x^2 + 9}$

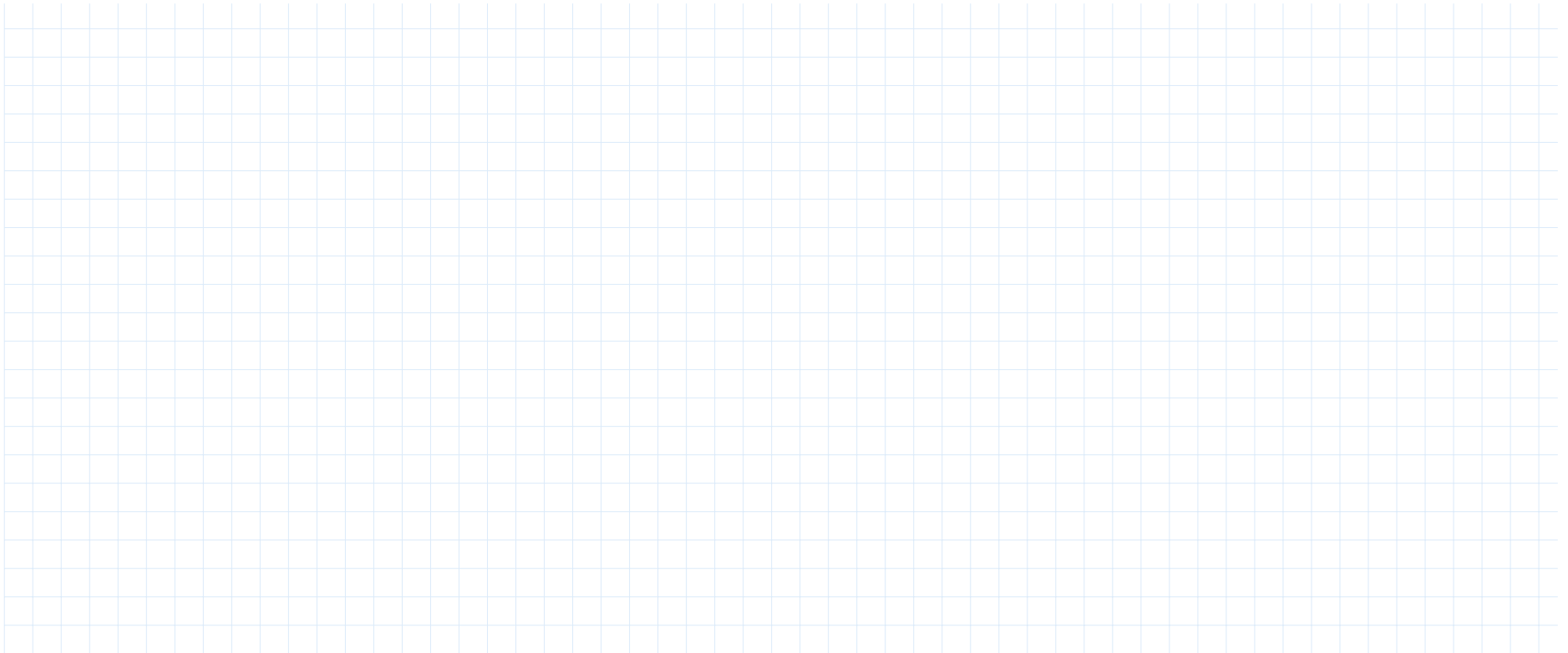


A Canceled Factor and a Hole

Let

$$f(x) = \frac{x^2 - 9}{x - 3}.$$

Find the domain, simplify the formula, locate the hole, and sketch the graph.

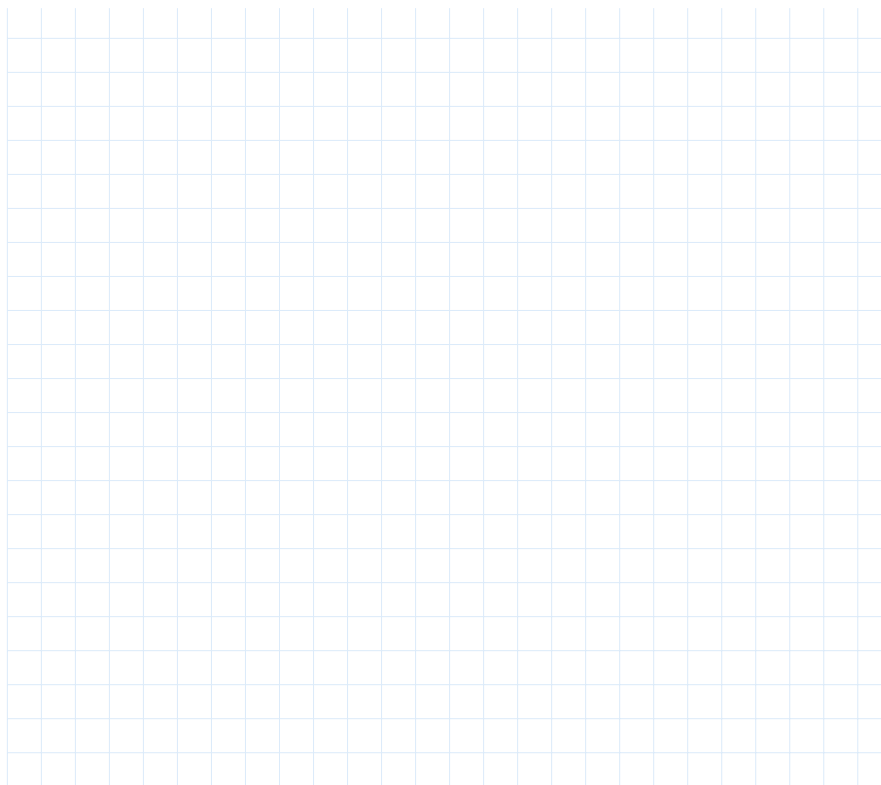


Two Elementary Reciprocal Functions

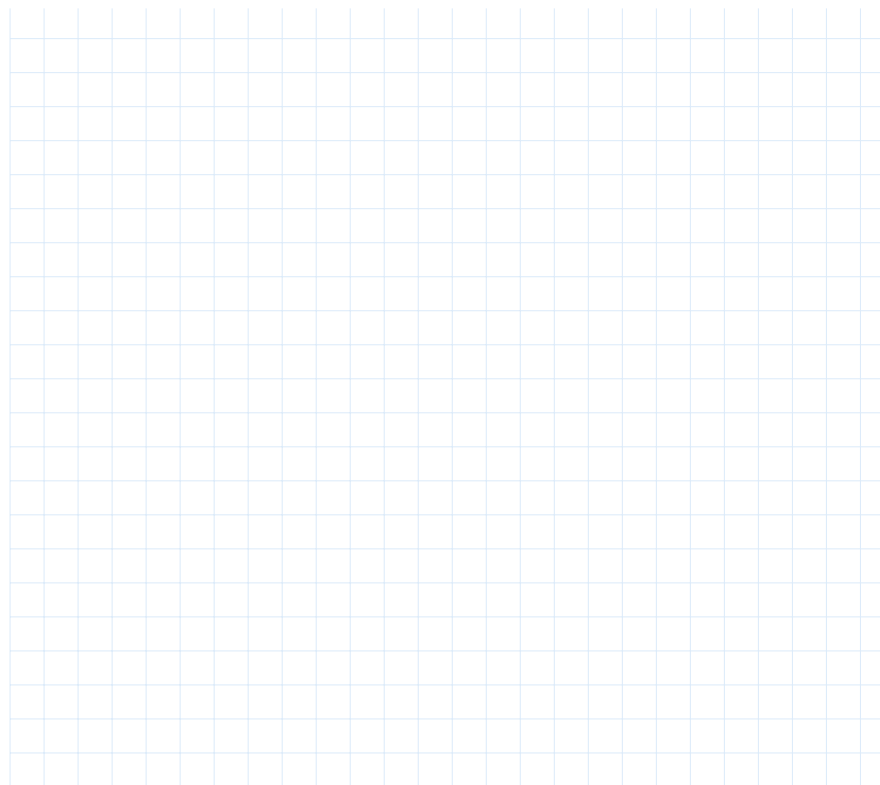
Section 3.4A

Sketch each function using a few points. State its domain and range. The lines $x = 0$ and $y = 0$ are its vertical and horizontal asymptotes.

$$y = \frac{1}{x}$$



$$y = \frac{1}{x^2}$$

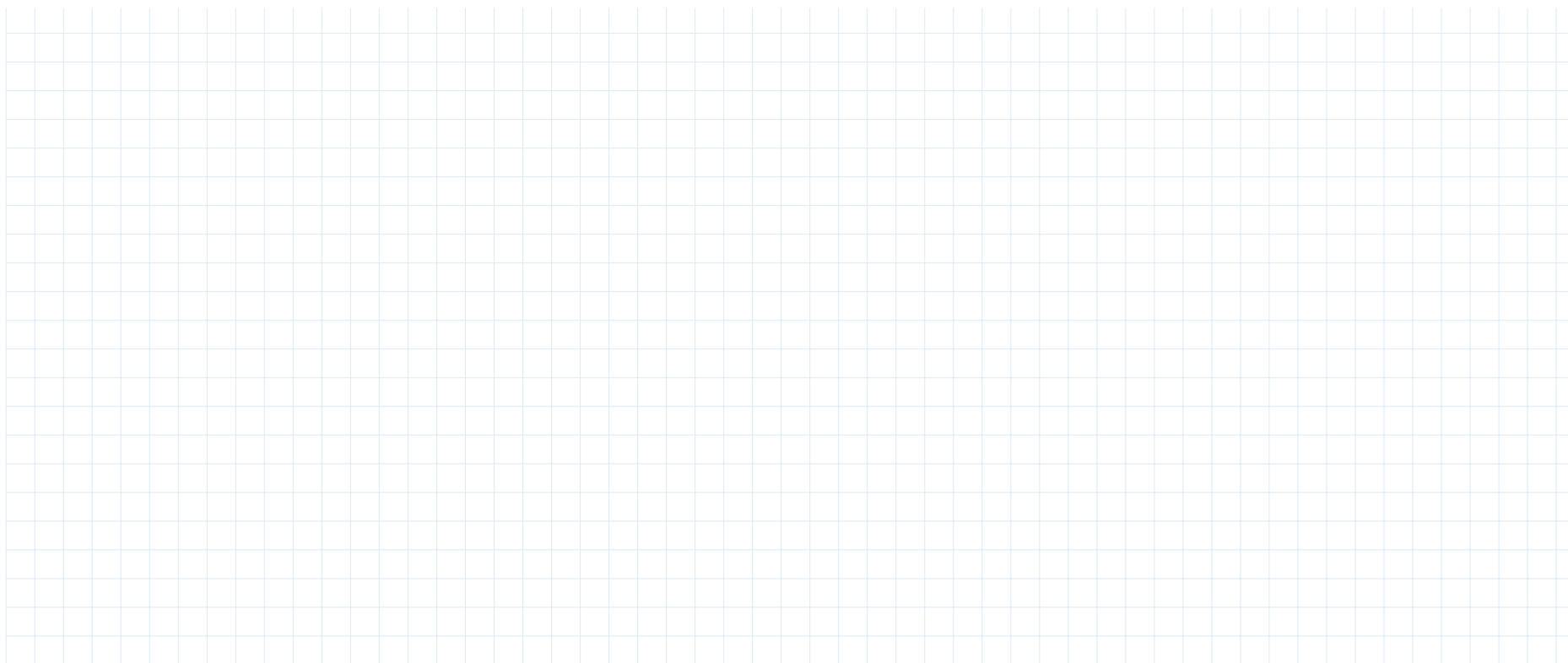


Transform a Reciprocal Function

Use the elementary function $y = \frac{1}{x}$ to sketch

$$g(x) = -\frac{2}{x-1} + 3.$$

Describe the transformations. State the domain, range, and both intercepts.



For

$$r(x) = \frac{x(x-2)}{x+1},$$

find the domain and all x - and y -intercepts. Explain why a zero of the denominator is not an x -intercept.

